

Nisarg Patel

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SUMMARY

Master of Science in Information Technology, GLS University | Full Stack Developer | Aspiring AI/ML & Software developer | [Portfolio](#) [GitHub](#) [LinkedIn](#)

SKILLS

Programming Languages: Java, Python, JavaScript

Frontend: React.js, HTML, CSS , TypeScript

Backend: Node.js, Express.js

Databases: MongoDB, MySQL, SQL

Tools: Git, GitHub

Cloud/Deployment: Render, Netlify, Cloudinary

PROJECTS

MedInsight AI – AI Medical Report Analyzer

[Live](#) [GitHub](#)

Built an AI-powered medical report analyzer that parses user-uploaded medical reports to extract biomarker data, providing a structured health overview and context-aware insights.

Engineered a custom parsing engine using pdf-parse (with layout-aware table extraction) and rule-based regex patterns to extract unstructured PDF text into organized biomarkers, validating results against standard reference ranges and persisting structured records in MongoDB.

Implemented a Retrieval-Augmented Generation (RAG) pipeline using MongoDB Atlas Vector Search chunking biomarkers by panel type, generating embeddings via Gemini's embedding model, and retrieving user/report-filtered context to ground chat responses and eliminate hallucinations.

Tech Stack: MongoDB, MongoDB Atlas Vector Search, Express.js, React.js, Node.js, Google Gemini API, Cloudinary, pdf-parse

Impact: Turns dense, complex medical reports into immediate, plain-language explanations of biomarkers, using RAG to ground every AI response in the user's own report data rather than generic LLM knowledge.

MealSnap – Smart Nutrition Assistant

[Live](#) [GitHub](#)

Contributed to an AI-driven nutrition assistant that processes grocery receipt photos using OCR, extracts food items, and matches them to databases to calculate nutrition profiles.

Developed a full-stack nutrition tracking platform featuring real-time macro insights, gap-based grocery recommendations, and daily intake logging.

Built OCR-to-nutrition workflows that parse scanned receipt text and automatically map it to a database using custom fuzzy matching logic with OCR error-tolerance.

Tech Stack: React.js, Python (FastAPI), SQLite (SQLAlchemy), OpenCV, EasyOCR

Impact: Replaces manual grocery logging by parsing entire shopping receipts from a single photo, returning automated nutritional breakdowns and recommendations in seconds.

OTHER AI/ML PROJECT

Hate Speech Detection using Transformer based NLP

[GitHub](#)

Data Preparation & NLP Pipeline – Combined five hate-speech datasets (*Davidson, OLID, HateXplain, Jigsaw, Civil Comments*) into a balanced 15K-sample, 3-class dataset (*Safe, Offensive, Hate Speech*). Applied spaCy NER masking, emoji demojization, and regex-based text normalization to reduce demographic bias.

Transformer Modeling & Evaluation – Fine-tuned DistilBERT using PyTorch with class-weighted loss to improve detection of minority hate-speech classes.

Achieved 85.5% Macro F1 and 95.7% ROC-AUC on a balanced 1,500-sample test set.

Tech Stack:

Machine Learning & NLP: Python, PyTorch, Hugging Face Transformers

Backend & API: FastAPI, REST APIs

Frontend: React, TypeScript, Tailwind CSS

ADDITIONAL

Proficient with modern IDEs and AI-assisted development tools.

Strong fundamentals in core IT subjects.

Worked on 10+ projects.

EDUCATION

Master of Science in Information Technology GLS University (2025- 2027)

Bachelor of Science in Information Technology GLS University (2022- 2025)

CERTIFICATIONS

Data Structures & Algorithms Apna College - Java programming fundamentals, Core data structures & algorithm, Problem-solving

Technology Job Simulation Deloitte Australia - Python-based software development, JSON data processing & analytical problem-solving

Seminar on Professional Skills GLS University - Professional communication & public speaking,Teamwork, collaboration

DISSERTATION / RESEARCH

A Study of Technologies for Building a Scalable and Secure Metaverse [View Paper](#)

Explored the core technologies, architecture, and security considerations required to build scalable, secure metaverse platforms.